A linear analysis of the Kelvin-Helmholtz instability

An exerpt from: A First Course in Magnetohydrodynamics, *a senior/graduate level text in preparation by D. Clarke.*

A *shear layer* is a narrow width of fluid (a layer) across which the flow speed along the layer changes significantly. Figure 1a shows a simplified shear layer from the "lab frame" in which the flow speed changes from $+v_0$ for $y > 0$ (the upper region 'u') to $-v_0$ for $y < 0$ (the lower region 'l') across an arbitrarily thin layer at $y = 0$. Left alone, an inviscid fluid would continue to flow like this in perpetuity, and thus a shear layer represents an equilibrium state in ideal hydrodynamics. However, equilibrium states need not be stable. In particular, if a shear layer is perturbed in the y-direction (as depicted in Fig. 1b), will the disturbance grow in time (an unstable system), or decay (a stable system)? If the disturbance grows, at what rate does it grow and what determines that rate?

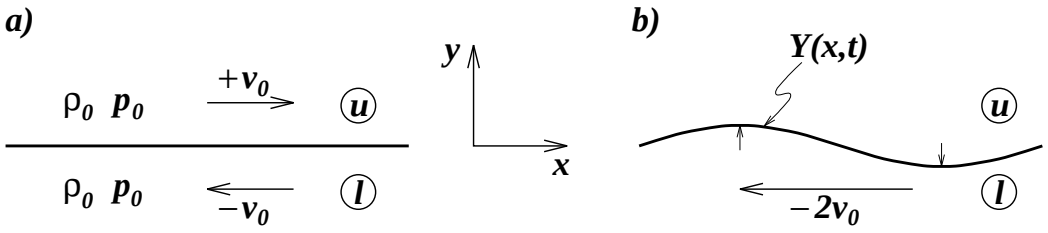


Figure 1: a) (left) A shear layer in an otherwise uniform fluid in the frame of reference of the lab. b) (right) The same shear layer from the frame of reference of the upper fluid (region 'u') with a sinusoidal perturbation applied to the shear layer.

We consider the time-evolution of a small-amplitude perturbation applied to a shear layer, and let $y = Y(x,t)$ describe the shape of the shear layer as a function of x and t. Since the distortions are small, we can use the *linearised* fluid equations from the co-moving frame of the fluid:

$$\frac{\partial p}{\partial t} = -\rho_0 c_s^2 \nabla \cdot \vec{v}; \tag{1}$$

$$\frac{\partial \vec{v}}{\partial t} = -\frac{1}{\rho_0} \nabla p, \tag{2}$$

where ρ_0 and c_s are the unperturbed density and sound speed respectively, and where p and v are the perturbed pressure and velocity. From this is it easy to show that in the rest frame of the fluid, the pressure is governed by a wave equation:

$$\frac{\partial^2 p}{\partial t^2} = c_s^2 \nabla^2 p. \tag{3}$$

Starting with the fluid in the *upper* stream ('u' in the figures), we seek *normal mode* solutions of the form:

$$p(x,y,t) \;=\; \mathcal{P}(y)\, e^{i(kx-\omega_u t)}, \tag{4}$$

where $\mathcal{P}(y)$ is the y-dependent amplitude of the perturbation, ω_u is the frequency of the normal mode as measured from the rest frame of region 'u', and k is the wave number, a Galilean invariant (independent of frame from which it is measured). Since the Fourier components of any general function are the normal modes, no loss of generality is suffered by assuming such a form for p.

$\mathcal{P}(y)$ is determined by substituting (4) into (3). To this end, we have:

$$\begin{aligned}
\frac{\partial^2 p}{\partial t^2} &= -\omega_u^2 \mathcal{P}(y)\, e^{i(kx-\omega_u t)};\\
\nabla p &= \left(ik\mathcal{P}(y), \frac{d\mathcal{P}(y)}{dy}, 0\right) e^{i(kx-\omega_u t)};\\
\nabla\cdot(\nabla p) \;=\; \nabla^2 p &= \left(-k^2\mathcal{P}(y) + \frac{d^2\mathcal{P}(y)}{dy^2}\right) e^{i(kx-\omega_u t)}.
\end{aligned}$$

Thus, (3) becomes:

$$\begin{aligned}
-\omega_u^2\, \mathcal{P}(y)\, e^{i(kx-\omega_u t)} &= c_s^2 \left(-k^2\mathcal{P}(y) + \frac{d^2\mathcal{P}(y)}{dy^2}\right) e^{i(kx-\omega_u t)}\\
\Rightarrow \quad \frac{d^2\mathcal{P}(y)}{dy^2} &= \left(k^2 - \frac{\omega_u^2}{c_s^2}\right)\mathcal{P}(y) \;\equiv\; \beta_u^2\,\mathcal{P}(y)\\
\Rightarrow \quad \mathcal{P}(y) &= A\, e^{\beta_u y} + B\, e^{-\beta_u y},
\end{aligned} \tag{5}$$

where A and B are constants of integration. In region 'u', $y > 0$ and thus $A = 0$ in order for $\mathcal{P}(y) \to 0$ as $y \to +\infty$. Thus,

$$\mathcal{P}(y) \;=\; B\, e^{-\beta_u y},$$

and (4) becomes:

$$p(x,y,t) \;=\; B\, e^{-\beta_u y}\, e^{i(kx-\omega_u t)}. \tag{6}$$

Next, the linearised Euler equation for v_y is

$$\frac{\partial v_y}{\partial t} \;=\; -\frac{1}{\rho_0}\frac{\partial p}{\partial y} \;=\; \frac{\beta_u}{\rho_0}\, B\, e^{-\beta_u y}\, e^{i(kx-\omega_u t)},$$

and since the x-component of the velocity, $v_{x,u}$, is zero in the rest frame of region 'u', we have at $y = 0$:

$$v_y \;=\; \frac{dY}{dt} \;=\; \frac{\partial Y}{\partial t} + v_{x,u}\frac{\partial Y}{\partial x} \;=\; \frac{\partial Y}{\partial t}.$$

Therefore:

$$\frac{\partial^2 Y}{\partial t^2} \;=\; -\frac{1}{\rho_0}\left.\frac{\partial p}{\partial y}\right|_{y=0} \;=\; \frac{\beta_u}{\rho_0}\, B\, e^{i(kx-\omega_u t)},$$

which can be integrated twice over time to get:

$$Y(x,t) \;=\; -\frac{\beta_u}{\rho_0\omega_u^2}\,B\,e^{i(kx-\omega_u t)}. \tag{7}$$

We can transform the expressions for p and Y [equations (6) and (7)] to the lab frame by the following simple argument. Perturbations with wave number k and thus wavelength $\lambda = 2\pi/k$ will have the same wavelength regardless of which frame of reference they are observed in (as already stated, k is a Galilean invariant). This is not true of the frequency. Suppose in the rest frame of region 'u', the waves are moving in the $+x$-direction, and suppose further that we observe $f = \omega_u/2\pi$ crests of the wave passing by in each second. Then, to shift to the lab frame as depicted in Fig. 1a, we must move with a velocity v_0 into the wavetrain and thus the number of crests we observe passing by per second must *increase*. Thus, transforming the above discussion to the lab frame is tantamount to applying a Doppler shift where the frequency observed in the lab frame, ω, is given by:

$$\omega \;=\; \omega_u + kv_0 \quad\Rightarrow\quad \omega_u \;=\; \omega - kv_0. \tag{8}$$

Substituting (8) into each of (6) and (7) yields:

$$p_>(x,y,t) \;=\; B\,e^{-\beta_u y}\,e^{i[kx-(\omega-kv_0)t]}, \tag{9}$$

$$Y_>(x,t) \;=\; -\frac{\beta_u}{\rho_0(\omega-kv_0)^2}\,B\,e^{i[kx-(\omega-kv_0)t]}. \tag{10}$$

where the subscript $_>$ indicates these expressions are valid for $y > 0$. Further, in terms of ω,

$$\beta_u \;=\; \sqrt{k^2 - \frac{(\omega-kv_0)^2}{c_s^2}}.$$

Expressions valid for lower region 'l' in Figs. 1a and 1b can be extracted directly from equations (9) and (10) simply by replacing v_0 with $-v_0$, and by noting that

$$\mathcal{P}(y) \;=\; A\,e^{+\beta_l y}; \qquad \beta_l \;=\; \sqrt{k^2 - \frac{(\omega+kv_0)^2}{c_s^2}}$$

in order for $\mathcal{P}(y) \to 0$ as $y \to -\infty$. Thus,

$$p_<(x,y,t) \;=\; A\,e^{\beta_l y}\,e^{i[kx-(\omega+kv_0)t]}, \tag{11}$$

$$Y_<(x,t) \;=\; \frac{\beta_l}{\rho_0(\omega+kv_0)^2}\,A\,e^{i[kx-(\omega+kv_0)t]}, \tag{12}$$

where the subscript $_<$ indicates these expressions are valid for $y < 0$.

To obtain the *dispersion relation* (how ω depends upon k), we exploit the singular property of p and Y that they must be continuous across the shear layer. Thus, at $y = 0$, we set $p_> = p_<$ and $Y_> = Y_<$ to get:

$$B\,e^{ikv_0 t} \;=\; A\,e^{-ikv_0 t} \tag{13}$$

$$-\frac{\beta_u}{(\omega-kv_0)^2}\,B\,e^{ikv_0 t} \;=\; \frac{\beta_l}{(\omega+kv_0)^2}\,A\,e^{-ikv_0 t}. \tag{14}$$

Dividing (14) by (13) yields:

$$\begin{aligned} -\beta_u(\omega + kv_0)^2 &= \beta_l(\omega - kv_0)^2 \\ \Rightarrow \quad \left(k^2 - \frac{(\omega - kv_0)^2}{c_s^2}\right)(\omega + kv_0)^4 &= \left(k^2 - \frac{(\omega + kv_0)^2}{c_s^2}\right)(\omega - kv_0)^4 \\ \Rightarrow \quad \left(\frac{\omega^2 - k^2v_0^2}{kc_s}\right)^2 &= 2\,(\omega^2 + k^2v_0^2), \end{aligned}$$

after some straight-forward algebra. Additional algebraic manipulations yield:

$$\left(\frac{\omega}{kc_s}\right)^4 - \left(\frac{\omega}{kc_s}\right)^2 2(M^2+1) + M^2(M^2-2) \;=\; 0,$$

where $M = v_0/c_s$ is the usual Mach number. The roots of the quadratic are:

$$\left(\frac{\omega}{kc_s}\right)^2 \;=\; M^2 + 1 \pm \sqrt{4M^2+1}$$

and thus four possible frequencies are admitted:

$$\begin{aligned} \omega_1 &= +kc_s\sqrt{M^2+1+\sqrt{4M^2+1}}; \\ \omega_2 &= -kc_s\sqrt{M^2+1+\sqrt{4M^2+1}}; \\ \omega_3 &= +kc_s\sqrt{M^2+1-\sqrt{4M^2+1}}; \\ \omega_4 &= -kc_s\sqrt{M^2+1-\sqrt{4M^2+1}}, \end{aligned}$$

where the perturbation, $Y(x,t)$, will develop as a linear combination of each of these four modes:

$$Y(x,t) \;=\; \sum_{j=1}^{4} \mathcal{C}_j(x,t)\, e^{-i\omega_j t}.$$

The first two frequencies, ω_1 and ω_2, are real and thus produce oscilliatory responses. The second two frequencies, ω_3 and ω_4, are real when:

$$M^2+1 \;>\; \sqrt{4M^2+1} \quad\Rightarrow\quad M^4+2M^2+1 \;>\; 4M^2+1 \quad\Rightarrow\quad M^4 \;>\; 2M^2 \quad\Rightarrow\quad M^2 \;>\; 2,$$

and thus when $M^2 > 2$, ω_3 and ω_4 produce oscillatory responses as well and the shear layer is stable to normal mode perturbations. On the other hand, for $0 < M^2 < 2$, ω_3 and ω_4 are imaginary, and we set:

$$\omega_3 \;=\; ikc_s\alpha; \qquad \omega_4 \;=\; -ikc_s\alpha,$$

where :

$$\alpha \;=\; \sqrt{\sqrt{4M^2+1} - M^2 - 1} \quad \in \mathcal{R} \;\text{ for } M^2 < 2. \tag{15}$$

Then,

$$\begin{aligned} e^{-i\omega_3 t} &= e^{\alpha k c_s t} \qquad \text{exponentially growing} \\ e^{-i\omega_4 t} &= e^{-\alpha k c_s t} \qquad \text{exponentially decaying} \end{aligned}$$

Thus, after some time t, the ω_3 term dominates all other terms, and the shear layer is unstable to normal mode perturbations. In this case,

$$Y(x,t) \;\propto\; e^{\alpha k c_s t},$$

which has a *growth-time*, τ, of:

$$\tau \;=\; \frac{1}{\alpha k c_s} \;=\; \frac{\lambda}{2\pi\alpha c_s}, \tag{16}$$

where λ is the wavelength of the normal mode. Evidently, the shorter the wavelength, the shorter the growth-time, and thus the faster the instability grows. It would seem that infinitely small wavelengths would grow infinitely fast, but this is not the case in practise. Viscous damping and/or surface tension along the shear layer preferentially damp out short-wavelength perturbations, and thus the very short wavelengths do not grow at all. But of the wavelengths not seriously affected by dissipative forces, equation (16) tells us the shorter ones will grow most rapidly.

Finally, we have seen that for $M^2 > 2$, the growth rate of the instability is the least (zero). At what Mach number is the growth rate the greatest? From (16), the shortest growth-time for a given wavelength corresponds to the greatest value for α, and thus of α^2. Therefore, from (15),

$$\frac{d(\alpha^2)}{d(M^2)} \;=\; \frac{4}{2\sqrt{4M^2+1}} - 1 = 0$$

$$\Rightarrow \quad \sqrt{4M^2+1} \;=\; 2 \quad \Rightarrow \quad 4M^2+1 \;=\; 4 \quad \Rightarrow \quad M^2 \;=\; 3/4.$$

Thus, the shear layer is most unstable when the Mach number for each of the upper and lower regions relative to the lab frame is $\sqrt{3}/2$, or when the relative Mach number between the layers is $\sqrt{3}$.